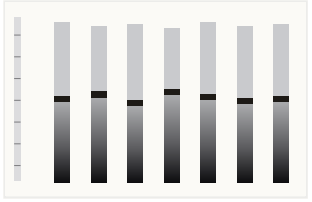
Automated detection of sediment rods and measurement of oxic-layer depth from photographs

A two-stage image-analysis pipeline converts a single overhead photograph of multiple sediment rods into a per-rod oxic-layer depth in millimetres. **Stage A** locates and crops each rod using Canny edge detection followed by orthogonal column and row projections of the edge map. **Stage B** measures, on each cropped rod, the vertical distance between a printed surface reference mark and the colour inflection that separates the oxic (oxidised) from the anoxic (reduced) zone.

STAGE A — Locate and crop individual rods from the tray photograph

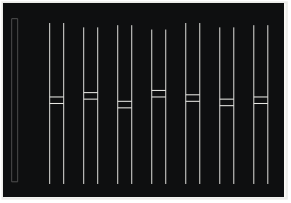
`input → N rod crops`

A1 — Original photograph



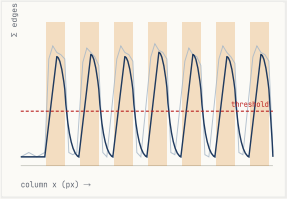
Overhead photograph of *N* rods on a tray with a length-calibration ruler. Tones encode oxidation: light silver above the surface mark, darkening to black below.

A2 — Edge map (grayscale → blur → Canny)



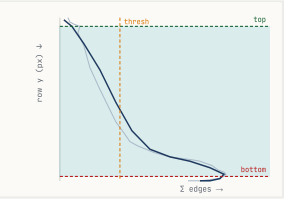
Convert to grayscale, apply a `5×5` Gaussian blur, then run Canny edge detection. The output is a binary map of intensity discontinuities — chiefly the vertical sides of each rod.

A3 — Column projection → rod x-bands



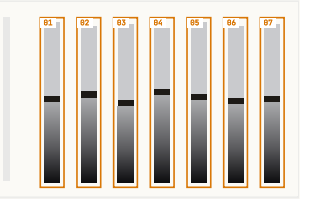
Sum the edge map down each column. Smooth, threshold against the median, and group contiguous active columns. Bands narrower than `8 px` or wider than `60 px` (e.g. the ruler) are rejected. Each accepted band gives one rod's $[x_0, x_1]$.

A4 — Row projection → rod y-extent



Sum the same edge map across each row. The first and last rows whose smoothed total exceeds threshold define a single `[y_top, y_bot]` applied to every rod.

A5 — Bounding boxes & per-rod crops



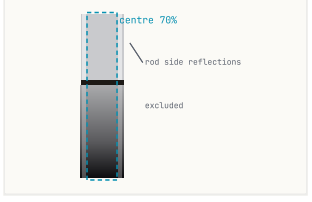
Combine each $[x_0, x_1]$ with the shared `[y_top, y_bot]`, pad slightly, and crop. Output: *N* labelled rod images written to `rod_crops/`.

`FOR EACH ROD CROP →`

STAGE B — Measure the oxic-layer depth on each cropped rod

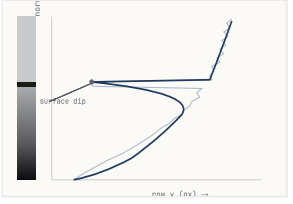
`1 rod crop → 1 depth (mm)`

B1 — Centre-strip extraction



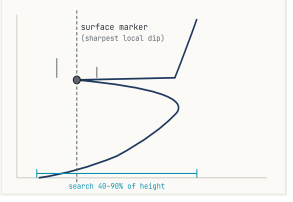
From each crop, keep the central `70 %` of the width. This removes specular highlights at the curved rod edges, which would otherwise bias the row-intensity profile.

B2 — Row-intensity profile



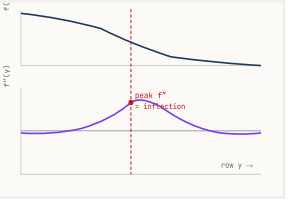
Sum pixel intensities across each row of the central strip and normalise to `[0, 1]`. Bright silver rows score high; the printed black mark and the dark anoxic tail score low.

B3 — Surface marker detection



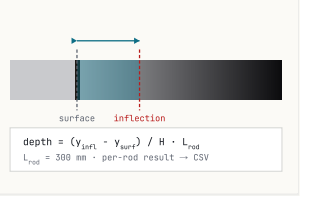
Scan a window across the lower half of the profile and pick the row with the largest local intensity dip — the printed reference mark that marks the sediment–water interface.

B4 — Inflection via Savitzky–Golay 2nd derivative



Below the surface marker, smooth the profile with a Savitzky–Golay filter and take its second derivative. The largest positive peak in `f''(y)` marks the colour inflection between oxic and anoxic sediment.

B5 — Oxic-layer depth (mm)



Pixel distance between surface and inflection is rescaled by the known total rod length to give the oxic-layer depth in millimetres. Results are aggregated across all rods into `results_summary.csv`.

LEGEND — detected rod band (column projection) · oxic layer extent · surface marker · inflection point · smoothed signal · raw signal